

Foundations of Multiple Testing

Part I ■ Error rates, closure, classical procedures

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Families and error rates

The testing family

Definition. A *family* is a set of null hypotheses $H_1, H_2, \dots, H_m$ that the researcher chooses to treat together for error-rate purposes.

Body fat example. Nine slope hypotheses in the regression of DEXfat on nine anthropometric predictors:

$$H_j : \beta_j = 0, \quad j = 1, \dots, 9.$$

The family is chosen, not given.

- A reader who pre-specified one row: family of size $m = 1$.
- A reader who scans the nine rows: family of size $m = 9$.
- A reader who ran many such regressions and reports the best: family larger still.

Test m hypotheses. Let m_0 be the number that are truly null, $m_1 = m - m_0$ the number that are truly false. After running the procedure, observe:

	Not rejected	Rejected	Total
True nulls	U	V	m_0
False nulls	T	S	m_1
Total	W	R	m

R is observable. V , S , U , T are unobservable random variables. m_0 , m_1 , m are fixed; m_0 and m_1 are unknown.

- V counts **false positives** (Type I errors).
- T counts **false negatives** (Type II errors).
- Error rates are built from functionals of V .

What drives the FWER?

FWER is driven by m_0 , the number of *true* nulls, not by m , the total number of tests.

$$\text{FWER} = P(V \geq 1) = P\left(\bigcup_{i: H_i \text{ true}} \{\text{reject } H_i\}\right).$$

This union grows with m_0 .

Situation	FWER (under independence)
All hypotheses false ($m_0 = 0$)	0 (no error possible)
Half true, half false ($m_0 = m/2$)	$1 - (1 - \alpha)^{m/2}$
All hypotheses true ($m_0 = m$)	$1 - (1 - \alpha)^m$

What you *report* does not shrink the family that *governs* your error rate.

Disjoint rejections vs. disjoint nulls

An easy confusion. These are *opposite* situations.

Disjoint rejection events.

The events $\{p_i \leq \alpha\}$ are mutually exclusive: at most one hypothesis can be rejected at a time.

Under the complete null, Bonferroni is exact:

$$P\left(\bigcup_i \{p_i \leq \alpha\}\right) = \sum_i P(p_i \leq \alpha) = m\alpha.$$

The union bound holds with equality. This is when the multiplicity problem is most severe relative to the joint test.

Disjoint null hypotheses.

At most one H_i can be true:

$$m_0 \leq 1.$$

$H_1: \theta < 0$, $H_2: \theta = 0$, $H_3: \theta > 0$.

Exactly one is true; the other two are false by construction.

If each test has size α , then

$$\text{FWER} \leq \alpha.$$

No correction is needed in this special case.

Disjoint rejections make Bonferroni exact and multiplicity most severe.

Disjoint nulls eliminate the multiplicity problem because $m_0 \leq 1$.

Four error rates

Per-comparison error rate.

$$\text{PCER} = \frac{E(V)}{m}$$

Average rate at which true nulls are falsely rejected. What unadjusted p -values control.

Familywise error rate.

$$\text{FWER} = P(V \geq 1)$$

Probability of *any* false rejection. What single-step max- t controls.

Generalized familywise error rate.

$$k\text{-FWER} = P(V \geq k)$$

Probability of k or more false rejections. A relaxation of FWER.

False discovery rate.

$$\text{FDR} = E\left[\frac{V}{R} \mid R \geq 1\right] P(R \geq 1)$$

Expected fraction of rejections that are false.

$$\begin{aligned} \text{PCER} &\leq \text{FWER}, & k\text{-FWER} &\leq \text{FWER for } k \geq 1 \\ \text{FDR} &\leq \text{FWER, with equality under the complete null.} \end{aligned}$$

The error rates are ordered

For any multiple-testing procedure,

$$\text{PCER} \leq \text{FDR} \leq \text{FWER}.$$

Under the complete null ($m_0 = m$), every rejection is a false rejection, so $\text{FDR} = \text{FWER}$.

complete null = global null

What this means.

- A procedure that controls FWER at level α also controls FDR and PCER at α .
- A procedure that controls FDR at level α need not control FWER.
- Controlling a more stringent rate tends to yield fewer rejections.

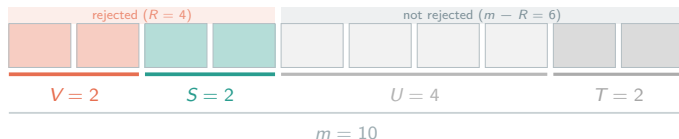
Practical consequence. The choice of error rate is a choice about how much evidence you demand. FWER is strict; FDR is more forgiving. Neither is universally correct.

Pick the error rate first. The procedure follows.

Error rates: one realisation, three sample quantities

Setup. $m = 10$ hypotheses, $m_0 = 6$ true nulls, $R = 4$ rejected, $V = 2$ false rejections.

false pos. (V) correct rej. (S) not rej. true null (U) missed (T)



$$\frac{V}{m}$$
$$\frac{2}{10} = 0.20$$

false rej. over *all* tests

$$\text{PCER} = E[V/m]$$

$$Q = V/R$$
$$\frac{2}{4} = 0.50$$

false rej. over *rejections* only

$$\text{FDR} = E[Q]$$

$$\mathbf{1}_{\{V \geq 1\}}$$
$$\mathbf{1}_{\{2 \geq 1\}} = 1$$

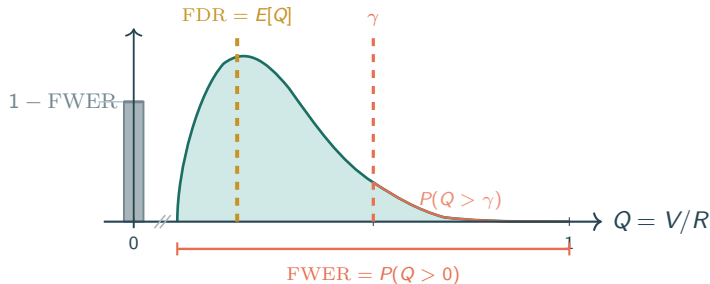
was *any* rejection false?

$$\text{FWER} = E[\mathbf{1}_{\{V \geq 1\}}]$$

$$\frac{V}{m} \leq \frac{V}{R} \leq \mathbf{1}_{\{V \geq 1\}} \text{ holds for every realisation } (m \geq R \text{ and } V/R \leq 1).$$

$$\text{Taking expectations: } \text{PCER} \leq \text{FDR} \leq \text{FWER}$$

The distribution of $Q = V/R$: false positives among rejections



Point mass at $Q = 0$
 $V = 0$: no false rejections.
Height = $1 - \text{FWER}$.

Continuous part on $(0, 1]$
Right-skewed. $\text{FDR} = E[Q]$ is the mean
of the whole mixed distribution.

Tail region $P(Q > \gamma)$
The shaded tail of the same density:
probability the false discovery fraction
exceeds γ .

FDR controlling procedures push $E[Q]$ left.
FWER controlling procedures shrink $P(V > 0) = P(Q > 0)$ to $\leq \alpha$.
Neither pins down the full shape of the distribution of Q .

Weak versus strong control

A procedure controls FWER in the **weak sense** if

$$P(V \geq 1 \mid H_0^{\text{global}}) \leq \alpha,$$

where $H_0^{\text{global}} = \bigcap_i H_i$ is the global null, under which all nulls are true.

A procedure controls FWER in the **strong sense** if

$$P(V \geq 1) \leq \alpha \quad \text{for all configurations of true and false nulls.}$$

Why the distinction matters.

- Weak control only protects you when *all* nulls are true.
- In practice, some nulls may be true and others false: strong control is what you usually want.
- Bonferroni, Holm, and single-step max- t all control FWER strongly.

Strong control is the default standard.
Always check which kind your procedure delivers.

Which error rate, and when?

Which rate, and when?

“To adjust or not to adjust, that is the question.” Or is it?

The real question is *which rate*, and sometimes the right rate requires no adjustment at all.

Examples from Hochberg & Tamhane (1987, Ch. 1). Each specifies a *decision structure*.

Each picks out a rate. The answers are not all the same, and that is the point.

Three families of answers we will see.

- **FWE**, strong control: one false rejection is one too many.
- **PFE** = $E(V)$: bound the expected number of false rejections.
- **PCE** = $E(V)/m$: each test stands alone; no joint error rate is controlled.

H&T notation: FWE = FWER, PCE = PCER. PFE = $E(V)$ is an upper bound on FWE, and the three rates satisfy $PCE \leq FWE \leq PFE$.

**The error rate follows from the decision structure.
Read the decision carefully, and the rate picks itself.**

Example 3.1: competing treatments with a control

Scenario. One of the main activities in a pharmaceutical company is the search for new drugs that are more effective than a standard drug or placebo.

Exploratory phase. In the exploratory phase it is common to start with a large number of candidate compounds. A screening device is needed to weed out potential noncontenders. The choice of error rate depends on the costs of confirmation, relative costs of Type I versus Type II errors, and prior experience.

Confirmatory phase. To obtain regulatory approval, the company must demonstrate by a confirmatory study that the new drug is at least as good as the control. Typically several drugs that have passed the exploratory screening phase are tested against the control. *No Type I error must be made in any of these comparisons*, because otherwise a drug that is actually inferior to the control may be recommended.

Example 3.1: the verdict

Exploratory phase.

The decision is a ranking and shortlisting. A false positive is costly but recoverable: the compound will fail later in confirmatory testing. The family is large; many null configurations are plausible because most compounds are inactive.

$$\text{PFE} = E(V)$$

Bound the expected number of false leads.

The level α is calibrated to the cost of confirmation.

Confirmatory phase.

The inferences feed a single, irreversible decision. A false positive means an inferior drug is approved and marketed. There is no subsequent safety net.

FWE, strong control

$P(V \geq 1) \leq \alpha$ under every configuration of true and false nulls.

Same scientific question, two phases, two rates.
The phase determines what kind of error is unacceptable.

Example 3.4: multiple outcome measures

Scenario. A new treatment is compared with a control on m pre-specified outcome measures.

Consider a confirmatory study where the list of measures is finite and specified in advance.

(e.g., $m = 5$: blood pressure, cholesterol, weight, heart rate, quality of life.)

The data, the statistics, and the p -values are the same regardless of which approval criterion is adopted. What differs is *what kind of error is unacceptable*.

Criterion A	Criterion B
The new treatment must do better on <i>all</i> measures.	The criterion is based on the <i>number</i> of measures on which the new treatment does better.
A single false positive corrupts the entire decision. ⇒ FWE strong control.	A single false positive need not corrupt the decision. ⇒ PFE or PCE.

Hochberg & Tamhane (1987), Example 3.4.

Example 3.4: the verdict

Criterion A: must win on all.

The conclusion requires *every* individual inference to be correct. A single false rejection can trigger a wrong recommendation.

The worst case occurs when exactly one null is true and is erroneously rejected, so strong FWE control is required.

FWE, strong control

$$P(V \geq 1) \leq \alpha.$$

Any false rejection corrupts the decision.

Criterion B: must win on enough.

The conclusion depends on a count of successes. Bounding the expected number of false wins directly bounds how much the count is inflated by chance.

FWE control would be unnecessarily harsh: it penalises risks the criterion does not face.

$$\text{PFE} = E(V)$$

Bound the expected number of false positives entering the count.

**The decision *rule*, not the number of tests, determines the rate.
Read the criterion carefully before choosing a procedure.**

Example 3.5: noncompeting treatments with a control

Scenario. A common control is included in the experiment for benchmark purposes, and the various treatments are combined in one experiment mainly for reasons of economy: they are *not competitive*.

For example, different school boards hire the same consulting organisation to evaluate their different educational programs. Each board is concerned only with the comparison of its own program with the control. A board's interpretation of the data is *unaffected* by the fact that other programs were included in the same experiment and compared with the same control.

The consulting organisation reports *separately* to each school board, not together as a group, and therefore has no special interest in ensuring that the FWE for the collection of all comparisons is controlled.

Here we have a simultaneous study but each inference is **autonomous** (although statistically dependent).

Example 3.5: the verdict

Why no family exists here. Each inference feeds only its own board's decision: no joint conclusion is drawn, and statistical dependence among the test statistics does not change that.

Feature	Present here?	Creates a family?
Tests run simultaneously	Yes	No
Test statistics are correlated	Yes	No
Results reported to the same organisation	Yes	No
A single joint decision is made	No	Yes

$$\text{PCE} = E(V)/m$$

Each inference runs at its own α . No multiplicity adjustment is warranted.

Statistical dependence $\neq$ a family.

A family exists when inferences feed *one* joint decision.

Five principles, distilled

1. **Define the family before seeing the data.** Include all inferences that *could* have been selected, not just those reported. Post-hoc family definition is circular.
2. **The error rate follows from the decision structure.** FWE when joint correctness is required; PFE when a bound on the false-positive count suffices; PCE only when inferences are truly autonomous.
3. **Stepwise procedures dominate single-step ones.** At equal FWER, stepwise procedures reject at least as much as their single-step counterparts. Holm dominates Bonferroni in terms of rejections at no additional cost, though Bonferroni has the advantage of directly yielding simultaneous confidence intervals.
4. **Exploit the joint distribution.** When test statistics are correlated, procedures that use the joint distribution (max- t , Westfall) gain power over procedures based on marginal p -values alone.
5. **There is no universally correct error rate.** Context, consequences, and research phase all matter. Pick deliberately.

The structure of a multiple test

Adjusted p -values

Definition.

$$p_i^{\text{adj}} = \inf\{\alpha : H_i \text{ is rejected by the procedure at level } \alpha\}.$$

What it buys you. A procedure controls FWER at level α if and only if

$$H_i \text{ is rejected} \iff p_i^{\text{adj}} \leq \alpha.$$

So one number per hypothesis; read the table as usual.

Body fat example. The adjusted- p column in the opener is p_i^{adj} for each row.

$p_{\text{kneebreadth}}^{\text{adj}} = 0.131$ means: at any level $\alpha < 0.131$, the procedure does not reject.

**Adjusted p -values move the multiplicity correction
from the critical value into the reported number.**

Single-step versus stepwise

Single-step. One threshold c_α for everyone. Every H_i is compared to the same critical value, derived under the *complete* null $\bigcap_i H_i$. Examples: Bonferroni, Šidák, single-step max- t .

Stepwise. Thresholds depend on which hypotheses have already been rejected. Derived under *reduced* nulls after sequential decisions.

- **Step-down:** start with most extreme statistic, work inward. Holm, step-down max- t .
- **Step-up:** start with least extreme statistic, work outward. Hochberg.

Why bother with stepwise? Stepwise procedures are uniformly at least as powerful as their single-step counterparts: every threshold is no bigger, and most are smaller. FWER control is preserved.

The union-intersection test

Setup. m null hypotheses $H_1, \dots, H_m$ with alternatives $K_i = \Theta \setminus H_i$.

Global null and alternative:

$$H = \bigcap_{i=1}^m H_i \quad \text{vs} \quad K = \bigcup_{i=1}^m K_i$$

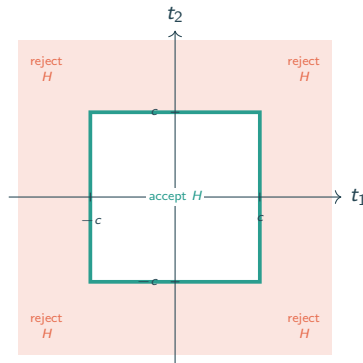
Construction. Reject H if *any* statistic is extreme:

$$\text{Reject } H \iff \max_i |t_i| > c$$

Acceptance region = $\bigcap_i \{|t_i| \leq c\}$: a *rectangle* in $(t_1, \dots, t_m)$.

Key limitation. Rejecting H tells us *something* is non-null; it does not say *which* H_i to reject.

Example. The body fat F -test overwhelmingly rejects $H: \beta_1 = \dots = \beta_9 = 0$ but gives no individual attribution.



$m = 2$, two-sided. Rejection = outside the box.

Max- t : the canonical union-intersection test

Test statistic.

$$t_{\max} = \max_{j=1, \dots, m} |t_j|$$

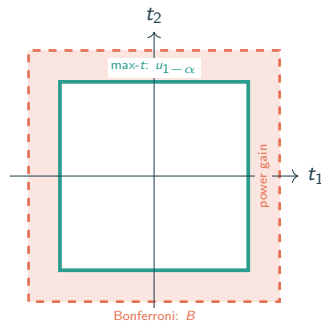
Reject H iff $t_{\max} > u_{1-\alpha}$, where $u_{1-\alpha}$ solves

$$P(t_{\max} \leq u_{1-\alpha} \mid H) = 1 - \alpha$$

Versus Bonferroni. Both use a box-shaped acceptance region. Bonferroni uses the marginal critical value $B = t_{1-\alpha/(2m), \nu}$. Max- t uses the joint-distribution quantile $u_{1-\alpha} \leq B$. Same shape, smaller box, uniformly more powerful.

Consonance. If $\max_j |t_j| > u_{1-\alpha}$, at least one $|t_j|$ exceeds $u_{1-\alpha}$: some H_j is immediately flagged.

The F -test lacks this property: it can declare that something is non-null without being able to point to which H_j is responsible.



**Same box, sharper cutoff.
Consonance gives in-
dividual attribution.**

From global to individual: the closure principle

The attribution problem. The UIT rejects $H = \bigcap H_i$ when $t_{\max}$ is large, but does not say *which* H_i is responsible. Naively rejecting all H_i with $|t_i| > c_\alpha$ does not give strong FWER control.

The closure principle (Marcus, Peritz & Gabriel 1976). Test *every intersection hypothesis*

$$H_I = \bigcap_{i \in I} H_i, \quad \emptyset \neq I \subseteq \{1, \dots, m\},$$

each at local level α . Reject H_i iff *every* H_I containing i is rejected.

Why it works. Falsely rejecting H_i requires first falsely rejecting H_{M_0} , the intersection of all true nulls. Since that test is valid at level α , $\text{FWER} \leq \alpha$ under any configuration.

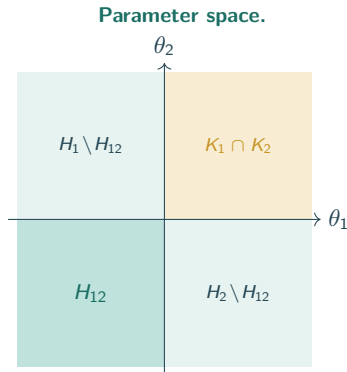
Two canonical instances.

- Holm = closure + Bonferroni on each H_I .
- Step-down max- t = closure + max- t on each H_I .

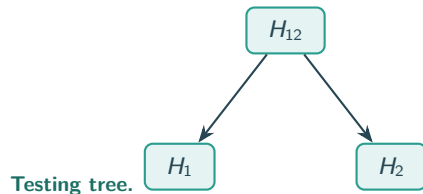
**Closure transforms a global test into individual tests
without inflating the error rate.**

Closure: two hypotheses

Setup. Two treatments vs. control: $\theta_i = \mu_i - \mu_0$, $H_i : \theta_i \leq 0$, $i = 1, 2$.



$H_{12} = H_1 \cap H_2$ (third quadrant).



Rejection rule. Each node tested at *full* level α .

H_1 rejected iff *both* H_{12} and H_1 reject at α .

H_2 rejected iff *both* H_{12} and H_2 reject at α .

Strong FWER control.

Whenever H_i is true, H_{12} is also true.

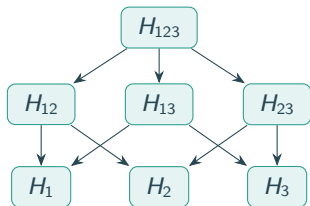
Closure forces H_{12} to be rejected before H_i , so:

$$P(V \geq 1) \leq P(H_{12} \text{ rejected}) \leq \alpha,$$

since $\{V \geq 1\} \subseteq \{H_{12} \text{ rejected}\}$.

Closure: three hypotheses

Closed family for $m = 3$: seven intersection hypotheses.



Each node tested at full level α .

The recipe.

1. Form $\tilde{\mathcal{H}} = \{H_I : \emptyset \neq I \subseteq \{1, \dots, m\}\}$.
2. Choose a local α -level test ϕ_I for each H_I .
3. Reject H_i iff ϕ_I rejects for every I containing i .
4. $\tilde{p}_i = \max_{I: i \in I} p_I$.

FWER control. For any M_0 : whenever H_i is true, H_{M_0} is also true and must be rejected first.

$$P(V \geq 1) \leq P(H_{M_0} \text{ rejected}) \leq \alpha.$$

Complexity. $2^m - 1$ tests: 7 for $m = 3$, 511 for $m = 9$. Holm and step-down max- t reach the same decisions in $O(m)$ steps.

The sequencing forces $\{V \geq 1\} \subseteq \{H_{M_0} \text{ rejected}\}$ by logic alone.
One valid α -level test at each node is all that is needed for strong FWER control.
The only question is: which test to use at each node?

Classical procedures

Procedure. Reject H_i if $p_i \leq \alpha/m$. Equivalently: $p_i^{\text{adj}} = \min(m \times p_i, 1)$.

Why it works. Union bound:

$$P\left(\bigcup_{i \in I_0} \{p_i \leq \alpha/m\}\right) \leq \sum_{i \in I_0} \frac{\alpha}{m} \leq \alpha$$

Strong FWER control. No distributional assumptions.

Price.

- Conservative under any positive dependence among the tests.
- Power loss grows with m .
- Ignores the joint distribution of the test statistics.

**Bonferroni is the safe choice that costs power.
Every other classical procedure is trying to buy some back.**

Holm (step-down)

Procedure. Sort p -values: $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$.

- Reject $H_{(1)}$ if $p_{(1)} \leq \alpha/m$. If not, stop.
- Reject $H_{(2)}$ if $p_{(2)} \leq \alpha/(m-1)$. If not, stop.
- ...
- Reject $H_{(i)}$ if $p_{(i)} \leq \alpha/(m-i+1)$. If not, stop.

Derivation. Holm is the closed test obtained by applying Bonferroni to each intersection H_I .

- At step i , the intersection of remaining hypotheses has size $m-i+1$.
- Bonferroni on that intersection uses threshold $\alpha/(m-i+1)$.
- Bonferroni itself is *not* a closed test: it uses the same threshold for every hypothesis regardless of what has already been rejected.

Adjusted p -values.

$$q_{(i)} = \min\{1, \max[(m-i+1)p_{(i)}, q_{(i-1)}]\}$$

The inner max ensures $q_{(i)} \geq q_{(i-1)}$: adjusted p -values must be non-decreasing. Without it, a later adjusted p could dip below an earlier one, contradicting the step-down logic.

The geometry of multiple testing

F-test versus Bonferroni: two acceptance regions

Test $H: \beta_1 = \beta_2 = 0$ in the linear model. In the (t_1, t_2) plane, the two procedures carve out different shapes.

F-test. Acceptance region:

$$t_1^2 - 2rt_1t_2 + t_2^2 \leq S^2(1 - r^2).$$

An *ellipse* whose shape depends on the correlation r .

At $r = 0$: a circle of radius S .

Bonferroni. Acceptance region:

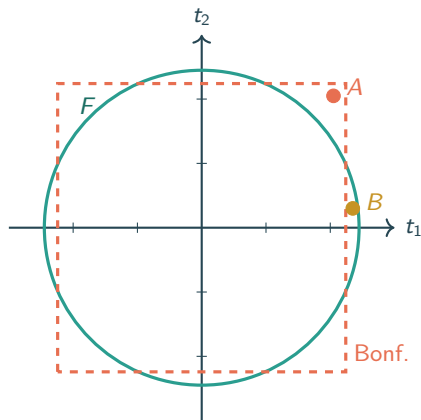
$$|t_1| \leq B \quad \text{and} \quad |t_2| \leq B.$$

A *square box* with half-side B , independent of r .

Quantity	$\alpha = 0.05, m = 2$
Ellipse cutoff S	$\sqrt{\chi_{2,.95}^2} = 2.448$ asymptotic
Box half-side B	$z_{1-\alpha/(2q)} = z_{.9875} = 2.241$

The two shapes agree in neither size nor geometry. *They make different decisions on the same data.*

Acceptance regions at $r = 0$



At $r = 0$, $\alpha = 0.05$:

- $S = 2.448$ (circle radius)
- $B = 2.241$ (box half-side)

Point A at (2.05, 2.05). Both t 's moderate.

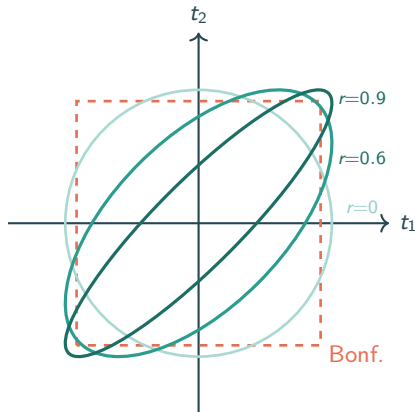
- $|t_i| = 2.05 < B \Rightarrow$ Bonf. accepts.
- $t_1^2 + t_2^2 = 8.40 > S^2 \Rightarrow$ F rejects.

Point B at (2.35, 0.30). One large, one small.

- $|t_1| = 2.35 > B \Rightarrow$ Bonf. rejects.
- $t_1^2 + t_2^2 = 5.61 < S^2 \Rightarrow$ F accepts.

Two points; the two tests disagree on both of them, *in opposite directions*.

Effect of correlation on the F-ellipse



As r grows, the F-ellipse becomes elongated along the diagonal $t_1 = t_2$.

What changes.

- F-ellipse widens along $t_1 = t_2$.
- F-ellipse narrows along $t_1 = -t_2$.
- Bonferroni box is fixed: its shape ignores r .

But Bonferroni's true size shrinks with $|r|$.

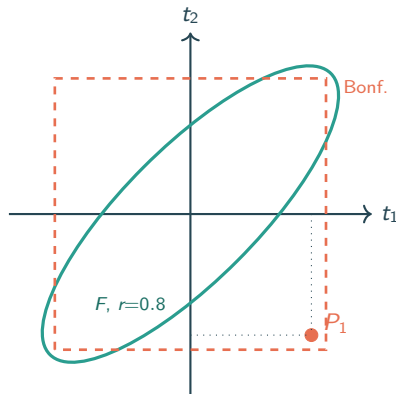
The box is fixed; the distribution is not. As r grows, less probability mass falls outside the box.

r	true size (nom. 0.05)
0.0	0.049
0.3	0.048
0.6	0.044
0.9	0.033

$$\text{true size} = 1 - P(-B \leq t_1 \leq B, -B \leq t_2 \leq B)$$

Individual t -stats cannot identify the source

F rejects, neither individual t reaches threshold. With $r = 0.8$, seeing t_1 and t_2 pull in opposite directions with equal force is unlikely under the null. F detects the joint pattern; Bonferroni misses it.



Point $P_1 = (2.0, -2.0)$, $r = 0.8$.

- $|t_1| = |t_2| = 2.0 < B = 2.24$: Bonf. accepts both.
- P_1 lies outside the F-ellipse: F rejects H .

F rejected the joint null but cannot say which H_i is responsible. Individual t -stats cannot answer that question either.

**Attribution requires the max- t test:
evaluate each t_j against
the *joint* distribution.**

Which test when?

A decision guide based on correlation structure and the question being asked.

Situation	Preference	Reason
Low $ r $, any direction	Either	Near-equal power.
High $ r $, effects opposite sign (anti-diagonal)	F, max- t	Ellipse is narrow there; F and max- t are sensitive to this pattern.
High $ r $, effects same sign (diagonal)	Bonferroni, Holm	Ellipse is wide along $t_1 = t_2$; Bonferroni box catches it instead.
Large m , sparse signal	Bonferroni, Holm	F power diluted across many df.
Need attribution: <i>which</i> H_i ?	Max- t	F rejects H globally only.

F tells you *whether*. Max- t tells you *which*.
Next: how to compute the joint critical value exactly.

Demo 1: Multiple testing adjustments on the body fat data

```
demo_bodyfat_classical_procs.R
```

- Nine slope hypotheses in the regression of DEXfat on anthropometric predictors.
- Raw p -values compared with Bonferroni and Holm adjustments.
- `kneebreadth`: significant raw ($p = 0.018$), not after adjustment. The marginal case where correction matters most.
- Holm dominates Bonferroni uniformly: same assumptions, never fewer rejections.

Strong signals survive every correction.

Weak signals are not rescued by any procedure.

The action is at the margin, and that is exactly where it matters.